HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Designed and implemented a **Python** data pipeline on **AWS Lambda** to process and transform ~5,000 real-time events per second, replacing a batch system and reducing data latency by **60%**.
- Developed and maintained **Java Spring Boot** REST APIs for core service integrations, handling approximately **1,500 requests per minute** during peak hours and ensuring robust error handling.
- Refactored complex relational database queries in **PostgreSQL**, improving query performance by an average of **30%** for critical reporting functions supporting enterprise systems.
- Collaborated with a cross-functional research team to co-design and iterate on algorithms that react in real-time to product feedback signals, directly impacting model adaptation speed.
- Optimized model inference pipelines by implementing techniques such as **quantization** and **knowledge distillation**, reducing model size by **40%** and improving inference latency by **25%**.
- Contributed to system-wide efficiency gains through cross-stack collaboration, focusing on optimizing the interaction between algorithms and real-time product data for enhanced user interaction.
- Containerized machine learning services using **Docker** and deployed them to an **AWS EKS** cluster, establishing basic CI/CD pipelines with **Jenkins** for streamlined updates and rollbacks.

SKILLS

Programming Languages – Python, Java, SQL, R

Frameworks & Libraries – Spring Boot, scikit-learn, pandas, NumPy, PyTorch, JAX, TensorFlow, Flask, FastAPI

Cloud & DevOps – AWS (EC2, S3, Lambda, CloudWatch, EKS), Docker, Kubernetes, Jenkins, Git

Databases – PostgreSQL, MySQL, Oracle

Data Analysis & BI – Power BI, Tableau, DAX, Power Query, Data Modelling, Data Warehousing

ML & Optimization – Model Efficiency, Real-time Alignment, Algorithmic Optimization, RLHF, Finetuning, Gradient Free Exploration

PROJECTS

Real-time ML Inference Service

- Engineered a **Python FastAPI** service to host a machine learning model for real-time data analysis, capable of processing ~2,000 requests per minute.
- Implemented model optimization techniques, including **quantization**, to reduce the model's memory footprint by **35%** and improve inference speed.
- Integrated the service with **PostgreSQL** for feature retrieval, ensuring low-latency data access for accurate model predictions.

Adaptive Learning System for Product Feedback

- Developed a **Python**-based system to capture and process user feedback from product interactions, feeding into an adaptive learning loop.
- Designed algorithms to analyze feedback in real-time, adjusting model parameters to improve performance by **15%** based on user interaction patterns.
- Deployed the system on **AWS Lambda**, ensuring scalability and efficient resource utilization for continuous adaptation.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)